

Synthesis and Characterization of Ultralong Hydroxyapatite Nanowire for Enhanced Osteochondral Regeneration

Summer Project Report
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Submitted by:
Arman Singh Jaryal
National Institute of Science Education and Research
(NISER), Bhubaneswar



Indian Institute of Technology, Mandi



Department of Biosciences and Bioengineering
Supervisor: Dr. Sumit Murab

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CERTIFICATE



This is to certify that Mr. **Arman Singh Jaryal** of the National Institute of Science Education and Research (NISER), Bhubaneswar has successfully completed his summer internship during the period of June-July 2024 on the topic of "**Synthesis and Characterization of Ultralong Hydroxyapatite Nanowire for Enhanced Osteochondral Regeneration**" at Translational Tissue Engineering Lab, Department of Biosciences and Bioengineering.

Indian Institute of Technology, Mandi

Dr. Sumit Murab
Supervisor
Translational Tissue Engineering Laboratory
Department of Biosciences and Bioengineering
Indian Institute of Technology, Mandi

DECLARATION

I hereby declare that I am the sole author of this **Report** in partial fulfillment of the requirements for a **Summer Internship Project** from Indian Institute of Technology (**IIT**), Mandi.

Signature of the Student

Date:

The project work reported in the report entitled "**Synthesis and Characterization of Ultralong Hydroxyapatite Nanowire for Enhanced Osteochondral Regeneration**" was carried out under my supervision, in the School of Biological Sciences and Bio-engineering at Indian Institute of Technology (**IIT**), Mandi, India.

Signature of the Guide/Supervisor

School:

Date:

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Abstract

In orthopedics, there are various challenges in treating diseases and damaged tissues. Although bone tissue possesses a natural regenerative capacity sufficient for healing small sites of damage, such as cracks and minor fractures, bone defects that exceed the critical size threshold (typically >2 cm, depending on the anatomical site) will take time to heal without external interventions.[7] Various ailments, such as osteoporosis, arthritis, and trauma-induced tissue damage, demand innovative solutions for the regeneration and restoration of orthopedic tissues. Metal bone-fixation devices often require subsequent surgical removal, the use of bone allografts is associated with a risk of disease transmission from the donor material, and the use of bone autografts results in additional morbidity associated with healing of the donor site. Tissue engineering has emerged as a promising approach, with 3D-printed orthopedic scaffolds gaining attention due to their potential to address these complex issues. 3D-printed scaffolds are valuable in bone regeneration due to their biodegradability and biocompatibility. However, using bioinks that combine biocompatible polymers with ceramic materials can further enhance their effectiveness. In this study, we synthesized Ultralong hydroxyapatite nanowires (HAPNW) and characterized them using Scanning Electron Microscopy (SEM), Attenuated Total Reflectance Fourier Transform Infrared Spectroscopy (ATR-FTIR), and X-ray Diffraction (XRD). Further Ag-Doping was carried on the sample for inducing antibacterial properties. The yield of HAPNW was determined to be used as a precursor in the synthesis of bio-ink. Although HAP nanowires have not yet been incorporated into bioink, the mechanical strength that HAP provides makes it a good bone biomimetic material that is also more rigid than a hydrogel-based ink.

Contents

Introduction	1
Objectives	3
Material and Methods	3
Experimental Procedure with Peanut Oil	4
Experimental Procedure with Oleic Acid	5
Variation in $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$ Concentration	5
Preparation of AgNP-incorporated HAPNWs	6
Preparation of AgNP-incorporated HAPNWs	6
Theoretical Yield Calculation of HAP Nanowires	8
Results and Discussion	10
Sample 1: Oleic Acid Precursor	11
Sample 2: Oleic Acid Precursor	12
Sample 3: Peanut Oil Precursor	13
EDX Calculations	16
Conclusion	17
References	17

Introduction

Bone is a complex tissue composed primarily of collagen fibers and inorganic mineral salts like hydroxyapatite, providing strength and rigidity. Bones' Macroscopic and microscopic structure level of osteons, nanostructural collagen fibrils, and mineral crystals, ensures that bones are both lightweight and extremely strong, capable of withstanding a wide range of stresses. It is necessary to understand the anatomy of bone for addressing orthopedic treatment challenges and developing new materials for bone regeneration. Bone tissue engineering, an innovative field, aims to develop biological substitutes to restore or improve bone function by combining principles from biology, engineering, and material science to create scaffolds that support new bone tissue growth. Scaffold-based techniques offer templates for cells to attach, proliferate, and differentiate, promoting effective tissue regeneration. Materials for scaffold fabrication include biocompatible and biodegradable polymers like polylactic acid (PLA), polyglycolic acid (PGA), and polycaprolactone (PCL); bioceramics like hydroxyapatite (HA) that enhance bone cell attachment and proliferation; and metals like titanium, known for their strength but limited by issues such as corrosion and lack of biodegradability.[6] 3D printing has revolutionized tissue engineering by enabling precise fabrication of complex scaffolds, allowing customization to patient-specific needs and creating intricate structures that enhance cell infiltration and nutrient delivery.[4] 3D bioprinting utilizes bioinks composed of living cells and biocompatible materials, employing techniques such as fused deposition modeling (FDM) to construct scaffolds layer by layer. Bioceramics, which include ceramics, glasses, and glass ceramics, are integral to bone tissue engineering due to their ability to bond with bone and release biologically active ions. Calcium phosphates like hydroxyapatite (HAP) and β -tricalcium phosphate are popular for mimicking the inorganic phase of bone, offering slow degradation and resorbable properties respectively. However, bioceramics' brittleness and susceptibility to fractures, especially under mechanical stress, limit their application in load-bearing areas. Additionally, the complex and costly manufacturing processes of bioceramics, along with their limited resorption capabilities, pose significant challenges for their

widespread use.

HAP nanowires (NW) are a promising material in bone tissue engineering due to their high surface area, bioactivity, and ability to release calcium and phosphate ions, essential for bone regeneration. Mixing HAP nanowires with alginate, a biopolymer derived from brown algae, can enhance the mechanical properties and bioactivity of scaffolds. Alginate forms hydrogels that support cell growth and differentiation. Chitosan and gelatin are also excellent biocompatible and degradable materials. Chitosan is derived from chitin found in crustacean shells, while gelatin is derived from collagen. Both support scaffold development by promoting cell attachment and growth.[1]

The use of Using HAP nanowires in these polymer matrices can significantly improve the scaffold's properties, making it suitable for bone regeneration. However, the brittleness of HAP nanowires and their complex synthesis pose challenges. Beyond bone tissue engineering, HAP nanowires are used in other industries due to their durability at high temperatures, biocompatibility, and wound healing properties, highlighting their versatility and potential for broader applications. In this project, we synthesized hydroxyapatite (HAP) nanowires and doped them with silver (Ag) for its antibacterial properties. We characterized the nanowires to achieve better morphology, creating longer and less brittle structures, which can be used in bioink synthesis. The ultralong HAP nanowires were synthesized by solvothermal/hydrothermal methods which include controlled chemical reactions to achieve high-purity nanomaterials. These advanced materials and fabrication techniques aim to address current orthopedic challenges, thereby improving bone healing and functionality. Ultralong Hydroxyapatite Nanowires, as defined in 2014, have a diameter less than 100 nm, length exceeding 100 μm , and an aspect ratio greater than 1000. [3] The high aspect ratios of these nanowires hold promise across various fields. Various synthesis methods include solvothermal, hydrothermal, microwave-assisted, rigid template, sol-gel hydrothermal, and reverse micelle. Challenges in their synthesis include achieving ultralong lengths and maintaining high flexibility, as current methods often produce shorter ($< 10 \mu\text{m}$) and more brittle nanowires.

Objectives

The objective of this study was to synthesize ultralong hydroxyapatite (HAP) nanowires with properties tailored for bioink formulations, focusing on achieving a replicable and efficient synthesis method. By employing peanut oil and oleic acid in the solvothermal synthesis process, we aimed to develop nanowires with enhanced morphology and reduced brittleness. This involved carefully combining calcium chloride, sodium hydroxide, sodium dihydrogen phosphate dihydrate, and deionized water, with systematic adjustments to precursor concentrations to optimize the Ca/P ratio. We also incorporated silver (Ag) doping to impart antibacterial properties, ensuring the nanowires not only matched the mineral phase of bone but also provided additional benefits for potential medical applications. The synthesis process was meticulously monitored, and the resulting nanowires were subjected to thorough characterization to analyze their structure and mechanical strength. A key objective was to confirm the replicability of our synthesis method, ensuring that the process could consistently produce high-quality nanowires. Additionally, we focused on optimizing the purification and derivation steps to enhance the efficiency and yield of ultralong HAP nanowires, making the process more feasible for large-scale production.[2] The study's ultimate goal was to produce a reliable and effective material that could significantly improve bioink formulations for bone tissue engineering applications.

Materials and Methods

For the synthesis of ultralong hydroxyapatite (HAP) nanowires, peanut oil (oleic acid content around 64%), calcium chloride anhydrous (CaCl_2), sodium hydroxide (NaOH), sodium dihydrogen phosphate dihydrate ($\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$), and deionized water were used. All chemical reagents were of analytical grade and used as received without further purification. Deionized water was used in all experiments where water was necessary

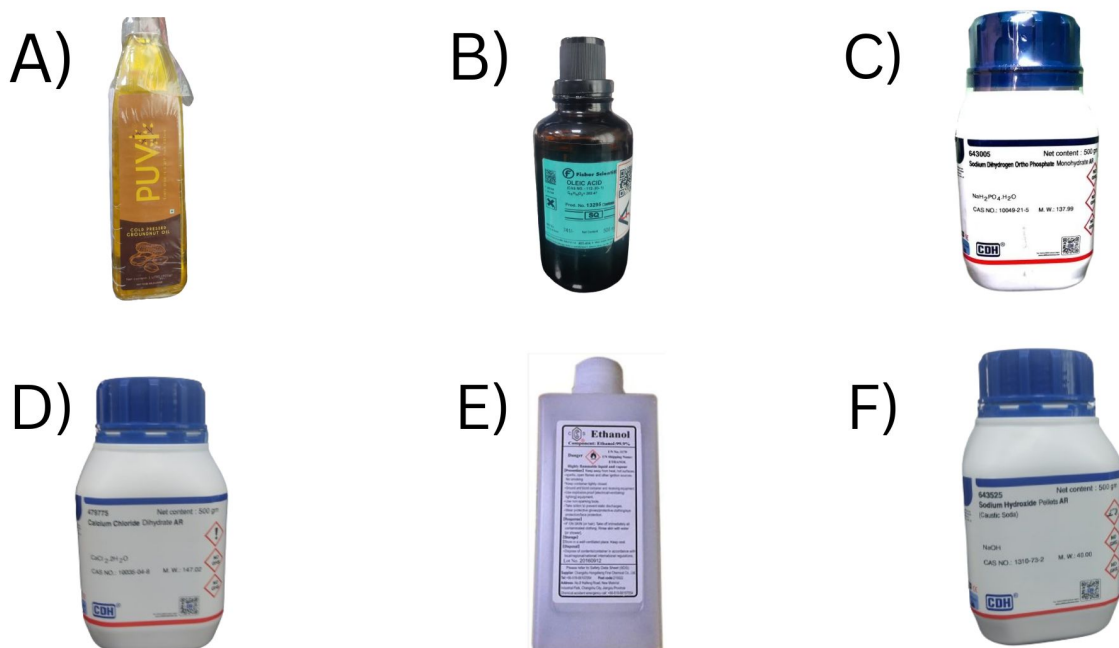


Fig 1: The mentioned are chemical reagents used in the respective experiments: A) Peanut Oil, B) Oleic Acid, C) $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$, D) Calcium Chloride (CaCl_2), E) Ethanol, F) Sodium Hydroxide (NaOH).

Experimental Procedure with Peanut Oil

For the solvothermal synthesis of Ultralong HAP nanowires using peanut oil, 9.360 g of peanut oil, 4.750 g of methanol, and 13.500 g of H_2O were mixed to form solution A.[3] Subsequently, 1.050 g of NaOH was dissolved in 15 mL of deionized water to form solution B, 0.166 g of CaCl_2 was dissolved in 15 mL of deionized water to form solution C, and 0.935 g of $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$ was dissolved in 15 mL of deionized water to form solution D. Solutions B, C, and D were sequentially added to solution A at 20-minute intervals under magnetic stirring. The resulting mixture was then poured into a 100 mL Teflon-lined stainless steel autoclave, sealed, and heat-treated at 180°C for 24 hours. After cooling to room temperature, the product was centrifuged and washed twice with deionized water and ethanol to obtain ultralong HAP nanowires at different RPM as shown in the Table 1.

Experimental Procedure with Oleic Acid

For the synthesis of Ultra long HAP nanowire using oleic acid, 0.330 g of CaCl_2 was dissolved in 15 mL of deionized water, forming solution E. Next, 1.050 g of NaOH was dissolved in 15 mL of deionized water to form solution F, and 2.779 g of $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$ was dissolved in 15 mL of deionized water to form solution G. Additionally, 10 mL of oleic acid was combined with 15 mL of ethanol under stirring to form solution H. Solution E was added dropwise to solution H under continuous stirring, followed by the gradual addition of solution F. Finally, solution G was slowly added to the mixture, bringing the final volume to approximately 70 mL. The mixture was stirred for 30 minutes, transferred to a Teflon-lined autoclave, and heated at 180°C for 24 hours.[5] The product was then allowed to cool naturally, centrifuged, and washed with deionized water and ethanol. It was then dried at 60°C for 12 hours to obtain the ultralong HAP nanowires.

Table 1: Sample Details with Precursor and Concentration Information

Sample	Precursor	Peanut Oil (mL)	Methanol (mL)	$\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$ (g)	CaCl_2 (g)	NaOH (g)	Ethanol (mL)	Oleic Acid (mL)
PO (Solution)	Peanut Oil	10.23	6	0.935	0.166	1.05	-	-
PO (Dried)	Peanut Oil	10.23	6	0.935	0.166	1.05	-	-
OA (Solution)	Oleic Acid	-	-	2.779	0.33	1.05	15	10
OA (Dried)	Oleic Acid	-	-	0.435	0.33	1.05	15	10
OA (Revised)	Oleic Acid	-	-	0.25	0.11	0.5	7.6	6.7

Variation in $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$ Concentration

In the third set of experiments, only the concentration of $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$ was changed. In the previous two experiments, the Ca/P ratio was maintained at 1:6 by varying the concentration of the Calcium and Phosphate precursors. The calculations for varying the concentration is given in the calculation section. Here, 0.330 g of CaCl_2 was dissolved in 15 mL of deionized water, 1.050 g of NaOH was dissolved in 15 mL of deionized water, and 0.435 g of $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$ was dissolved in 15 mL of deionized water. These solutions were combined with 10 mL of oleic acid and 15 mL of ethanol under the same conditions as previously described. The subsequent steps of stirring, transferring to the autoclave, heating, cooling, centrifuging, washing, and drying were performed as before to obtain the ultralong HAP nanowires.

In all experiments, the final ultralong HAP nanowires were obtained through the same rigorous process of preparation, reaction, and purification, ensuring consistency and reliability in the resultant materials.

Table 2: Centrifugation and Washing Details of the Samples

Sample	C1 (12 min)	C2 (10 min)	C3 (8 min)	Sonication	C4 (7 min)	C5 (5 min)	C6 (5 min)
PO (Solution)	6000	6000	-	-	-	-	-
PO (Dried)	6000	6000	6000	-	-	-	-
OA (Solution)	6000	6000	6000	-	-	-	-
OA (Dried)	6000	6000	4500	-	-	-	-
OA (Revised)	4000	4000	4000	5 min	2500	2500	2500

The sample was washed with deionized water first, followed by ethanol to remove excess oil from the sample. Once the supernatant is clear and free of oil, the ethanol is discarded, and the pellet is kept at room temperature to air dry. To remove excess oil and HAP entanglement, we sonicated the revised sample.

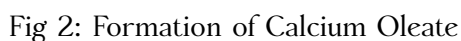
Preparation of AgNP-incorporated HAPNWs

For loading with AgNPs, 70mg of HAPNWs and 21mg of AgNO_3 were dispersed in 35mL of deionized water, and the resulting suspension was stirred for 3 hours in the sunlight at room temperature. The product (HAPNWs@AgNP nanowires) is assumed to be made after filtration and washing twice with deionized water, but no characterization has been performed to support this claim.[8] The product was grey in color, unlike the white coloured HAP nanowire product, indicating a successful reaction of AgNO_3 and HAP.

Role and Mechanism of Oleic Acid in HAP Nanowire Synthesis

Oleic acid is crucial in the synthesis of hydroxyapatite (HAP) nanowires. It forms a calcium-oleate complex and releases Ca ions uniformly. Oleic acid coats nanowires and prevents clumping for high purity. It reduces surface tension and stabilizes one-dimensional structures. Oleic acid is a mono-unsaturated omega-9 fatty acid, because the single double bond is on the carbon nine position. Oleic acid has the formula

Formation of Calcium Oleate

$$2\text{C}_{18}\text{H}_{34}\text{O}_2 + \text{Ca}^{2+} \rightarrow \text{Ca}(\text{C}_{18}\text{H}_{33}\text{COO})_2 + 2\text{H}^+$$


7

Controlled Release of Calcium Ions

The calcium oleate formed in the previous step acts as a calcium ion reservoir. Over time, calcium oleate releases calcium ions (Ca^{2+}) in a controlled manner, ensuring a steady supply of calcium ions for the formation of HAP.



Nucleation and Growth of HAP

The released calcium ions react with phosphate ions (PO_4^{3-}) and hydroxide ions (OH^-) to form hydroxyapatite nanowires. The overall reaction is:



Theoretical Yield Calculation of HAP Nanowires

The following steps outline the calculation of the theoretical yield of Hydroxyapatite (HAP) nanowires:

Determine the Moles of Each Reactant

Calcium chloride (CaCl_2):

$$\text{Molar Mass of CaCl}_2 = 40.08(\text{Ca}) + 2 \times 35.45(\text{Cl}) = 110.98 \text{ g/mol}$$

$$\text{Mass of CaCl}_2 = 0.330 \text{ g}$$

$$\text{Moles of CaCl}_2 = \frac{0.330 \text{ g}}{110.98 \text{ g/mol}} = 0.00297 \text{ moles}$$

Sodium hydroxide (NaOH):

$$\text{Molar Mass of NaOH} = 22.99(\text{Na}) + 15.999(\text{O}) + 1.008(\text{H}) = 39.997 \text{ g/mol}$$

$$\text{Mass of NaOH} = 1.05 \text{ g}$$

$$\text{Moles of NaOH} = \frac{1.05 \text{ g}}{39.997 \text{ g/mol}} = 0.02625 \text{ moles}$$

Sodium dihydrogen phosphate dihydrate ($\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$):

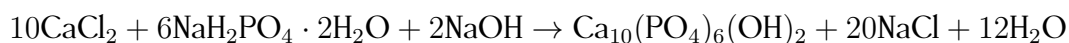
$$\begin{aligned}\text{Molar Mass of } \text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O} &= 22.99(\text{Na}) + 2 \times 1.008(\text{H}) + 30.97(\text{P}) + 4 \times 15.999(\text{O}) + 2 \times \\ &(2 \times 1.008 + 15.999) = 156.01 \text{ g/mol}\end{aligned}$$

$$\text{Mass of } \text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O} = 2.779 \text{ g}$$

$$\text{Moles of } \text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O} = \frac{2.779 \text{ g}}{156.01 \text{ g/mol}} = 0.01781 \text{ moles}$$

Identify the Limiting Reagent

The balanced chemical equation for the formation of Hydroxyapatite (HAP) is:



From the equation, 10 moles of CaCl_2 react with 6 moles of $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$ and 2 moles of NaOH to form 1 mole of HAP.

Mole Ratios Based on the Balanced Equation:

$$\begin{aligned}\text{For } \text{CaCl}_2 : \frac{10}{10} &= 1 \\ \text{For } \text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O} : \frac{6}{10} &= 0.6 \\ \text{For } \text{NaOH} : \frac{2}{10} &= 0.2\end{aligned}$$

Actual Mole Ratios Relative to CaCl_2 :

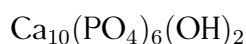
$$\begin{aligned}\text{Ratio of } \text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O} \text{ to } \text{CaCl}_2 : \frac{0.01781}{0.00297} &\approx 5.99 \\ \text{Ratio of } \text{NaOH} \text{ to } \text{CaCl}_2 : \frac{0.02625}{0.00297} &\approx 8.83\end{aligned}$$

Since the actual mole ratios of NaOH and $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$ are greater than the stoichiometric ratios, CaCl_2 is the limiting reagent.

Calculate the Moles of HAP Formed

$$\text{Moles of HAP} = \frac{0.00297 \text{ moles } \text{CaCl}_2}{10} = 0.000297 \text{ moles}$$

Calculate the Molar Mass of HAP



$$\begin{aligned}\text{Molar Mass} &= 10 \times 40.08 + 6 \times (30.97 + 4 \times 15.999) + 2 \times (1.008 + 15.999) \\ &\approx 1004.64 \text{ g/mol}\end{aligned}$$

Calculate the Theoretical Yield in Grams

$$\text{Theoretical Yield} = 0.000297 \text{ moles} \times 1004.64 \text{ g/mol} \approx 0.298 \text{ g}$$

$$0.298 \text{ g} \times 1000 = 298 \text{ mg}$$

The theoretical yield of HAP nanowires is approximately 298 mg. This calculation is essential for determining the efficiency and feasibility of the synthesis process in order to figure out the amount of nanowire going to be used in the bioink synthesis.

Results and Discussion

To establish the successful synthesis of ultralong HAP nanowires, it is crucial to examine their surface morphology and structural characteristics. The morphology of the synthesized nanowires provides insights into their physical dimensions, uniformity, and overall structural integrity, which are essential for determining their suitability for biomedical applications. In particular, parameters such as diameter, length, and aspect ratio significantly influence the nanowires' mechanical properties and biological interactions.

In this study, the surface morphology of the synthesized ultralong HAP nanowires was analyzed using scanning electron microscopy (SEM). As depicted in Fig. 3, the SEM micrographs of the HAP nanowires synthesized via the calcium oleate precursor solvothermal method at a Ca/P molar ratio of 1.54 and 180 °C for 24 hours reveal the formation of nanowires with an average diameter of 47.96 nm. This uniformity in diameter is indicative of a controlled synthesis process, crucial for ensuring consistent performance in potential applications.

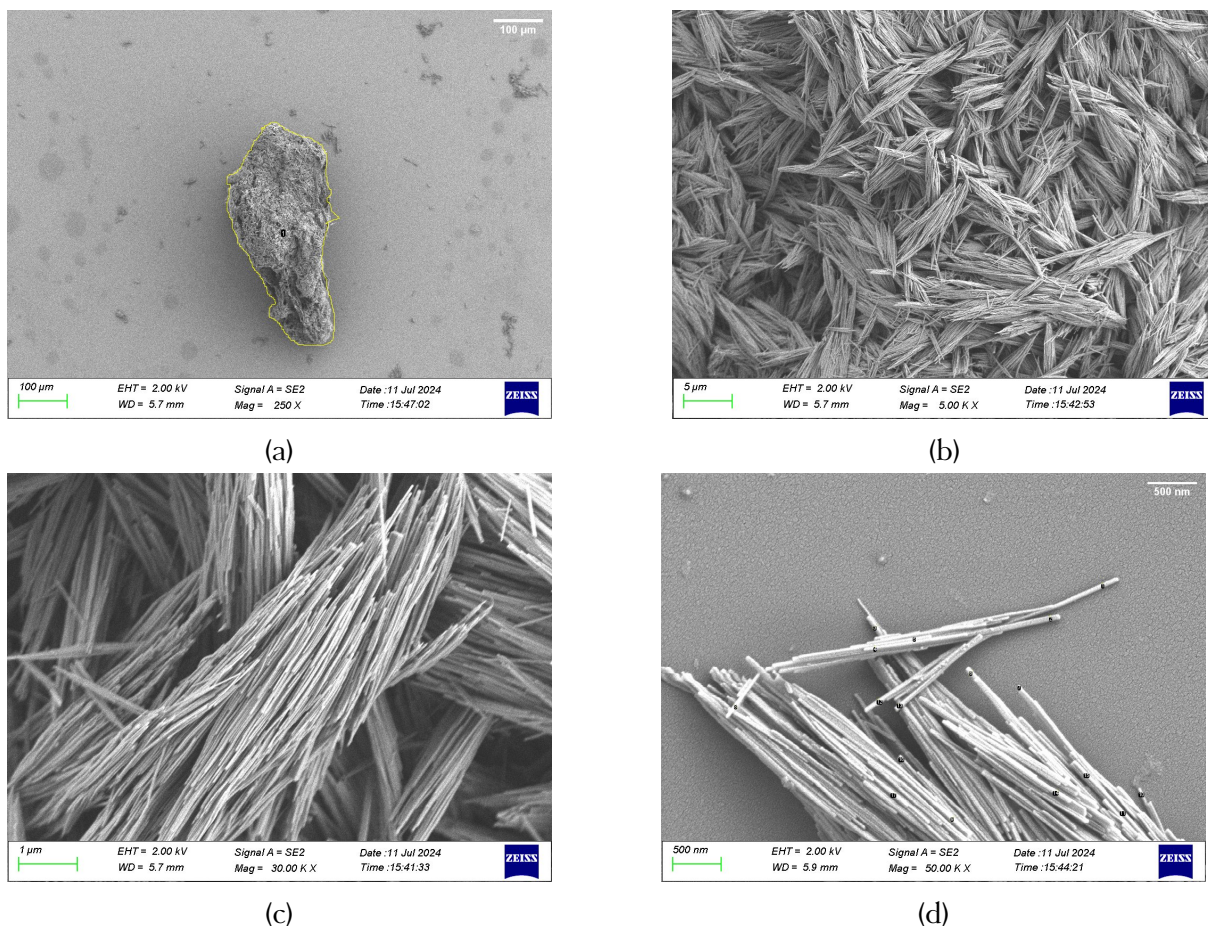


Fig 3: (a) The Clump was observed at $100\mu\text{m}$ consists of miniscule Nanowire, (b) The Nanowires were observed entangled at $5\mu\text{m}$, (c) The Nanowire was observed at $1\mu\text{m}$ (d) A Few individual Nanowire was observed at 500nm

Fourier-transform infrared spectroscopy (FTIR) was performed to identify the functional groups present in the nanowires. The Fourier transform infrared (FTIR) spectrum (Fig 5) indicates the interaction between carboxylic groups of oleic acid molecules and HAP nanowires.

Sample 1: Oleic Acid Precursor

The FTIR data of Sample 1, synthesized with an oleic acid precursor, show several characteristic peaks. The peak at 3568cm^{-1} can be assigned to the hydroxyl group of HAP. Two sharp bands at 2922cm^{-1} and 2854cm^{-1} are attributed to the asymmetric and symmetric stretches of the CH_2 groups in oleic acid molecules, respectively. The peak at 1711cm^{-1} corresponds to the $\text{C}=\text{O}$ stretch in oleic acid. The absorption bands at 1553cm^{-1} and 1462cm^{-1} are attributed to the asymmetric and symmetric stretches

of the COO^- group, respectively. The band at 1088 cm^{-1} is associated with the PO_4^{3-} group. Additionally, the bands corresponding to the PO_4^{3-} group are observed at about 1017 cm^{-1} , 599 cm^{-1} , and 558 cm^{-1} . According to the literature, the absorption bands of oleic acid are located at about 1286 cm^{-1} , 1417 cm^{-1} , 1460 cm^{-1} , and 1710 cm^{-1} in the FTIR spectrum.

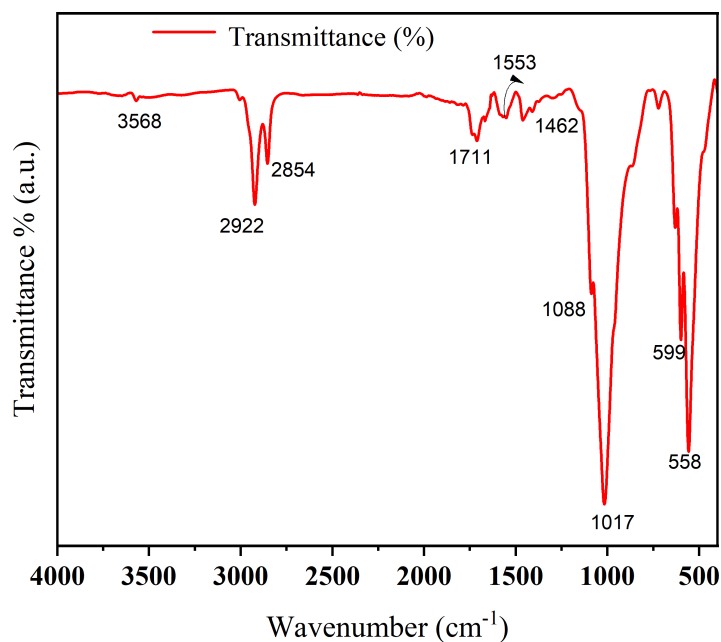


Fig 4: The FTIR data of Sample 1, synthesized with an oleic acid precursor

Sample 2: Oleic Acid Precursor

For Sample 2, also synthesized with an oleic acid precursor, the FTIR spectrum shows similar characteristic peaks. The peak at 3571 cm^{-1} can be assigned to the hydroxyl group of HAP. The bands at 2921 cm^{-1} and 2852 cm^{-1} correspond to the asymmetric and symmetric stretches of the CH_2 groups in oleic acid molecules, respectively. The peak at 1711 cm^{-1} is attributed to the C=O stretch in oleic acid. The absorption bands at 1557 cm^{-1} and 1447 cm^{-1} are due to the asymmetric and symmetric stretches of the COO^- group, respectively. The band at 1088 cm^{-1} is associated with the PO_4^{3-} group. Additionally, the bands corresponding to the PO_4^{3-} group are observed at about 1018 cm^{-1} .

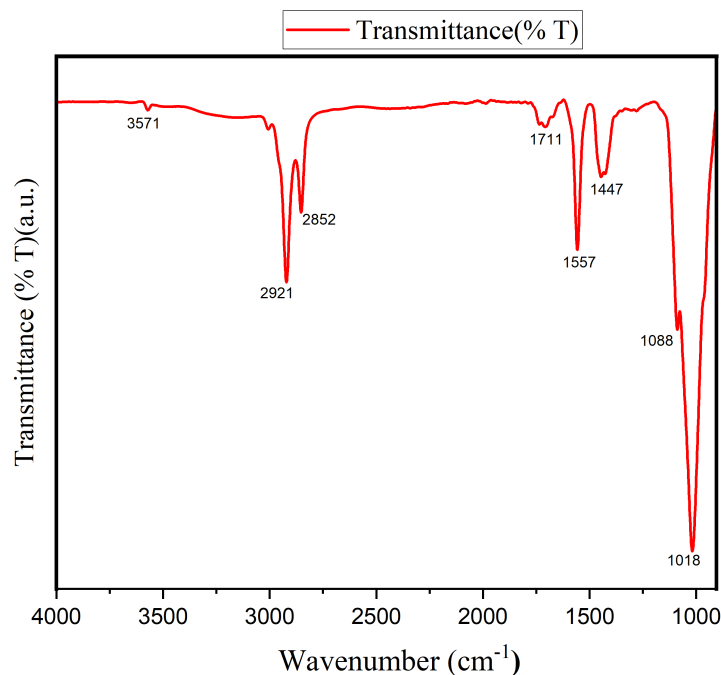


Fig 5: The FTIR data of Sample 2, synthesized with an oleic acid precursor with a variation in $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$ Concentration

Sample 3: Peanut Oil Precursor

The FTIR data of Sample 3, synthesized with a peanut oil precursor, exhibit distinct peaks. The peak at 3370 cm^{-1} can be attributed to the hydroxyl group of HAP. The peaks at 3122 cm^{-1} and 2916 cm^{-1} correspond to the asymmetric stretches of the CH_2 groups. The peak at 2850 cm^{-1} is associated with the symmetric stretch of the CH_2 groups. The peak at 1709 cm^{-1} corresponds to the $\text{C}=\text{O}$ stretch. The absorption bands at 1467 cm^{-1} and 1415 cm^{-1} are due to the asymmetric and symmetric stretches of the COO^- group, respectively. The band at 1256 cm^{-1} is related to the PO_4^{3-} group. Additionally, the bands at 1047 cm^{-1} , 942 cm^{-1} , 860 cm^{-1} , 605 cm^{-1} , and 507 cm^{-1} correspond to the PO_4^{3-} group.

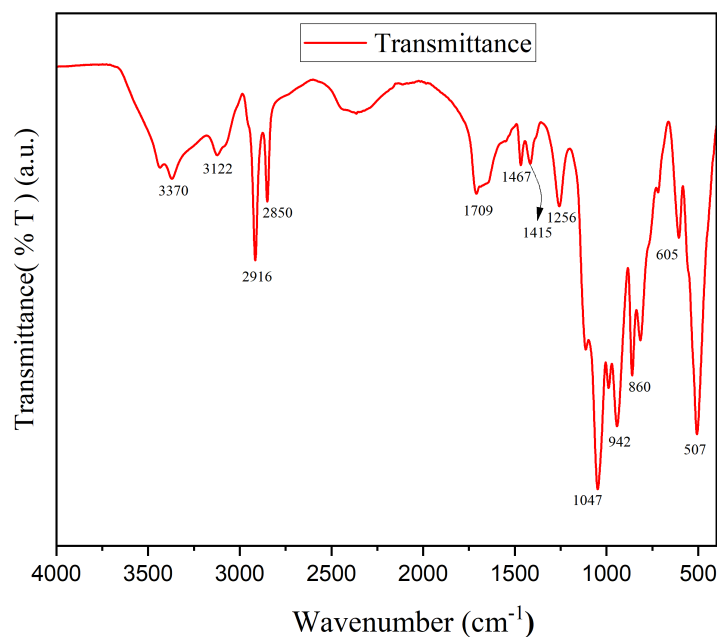


Fig 6: The FTIR data of Sample 3, synthesized with an Peanut Oil precursor

The differences in the FTIR spectra among Samples 1, 2, and 3 can be attributed to the different precursors used in their synthesis. The oleic acid precursor results in characteristic peaks associated with oleic acid functional groups, while the peanut oil precursor leads to distinct peaks indicative of its impurity.

The samples of ultralong HAP nanowires synthesized by the calcium oleate precursor solvothermal method using Oleic acid further characterized by X-ray diffraction (XRD). The obtained XRD patterns and FTIR spectra are shown in Fig 7.

The XRD patterns of these three samples (Fig. 4) can be indexed to the single crystal phase of hydroxyapatite with a hexagonal structure with a space group of $P6_3/m$ (176), $\alpha = \beta = 90^\circ$, and $\gamma = 120^\circ$ (JCPDS No. 09-0432). The sharp and well-defined peaks in the XRD patterns indicate the high crystallinity of the synthesized HAP nanowires.

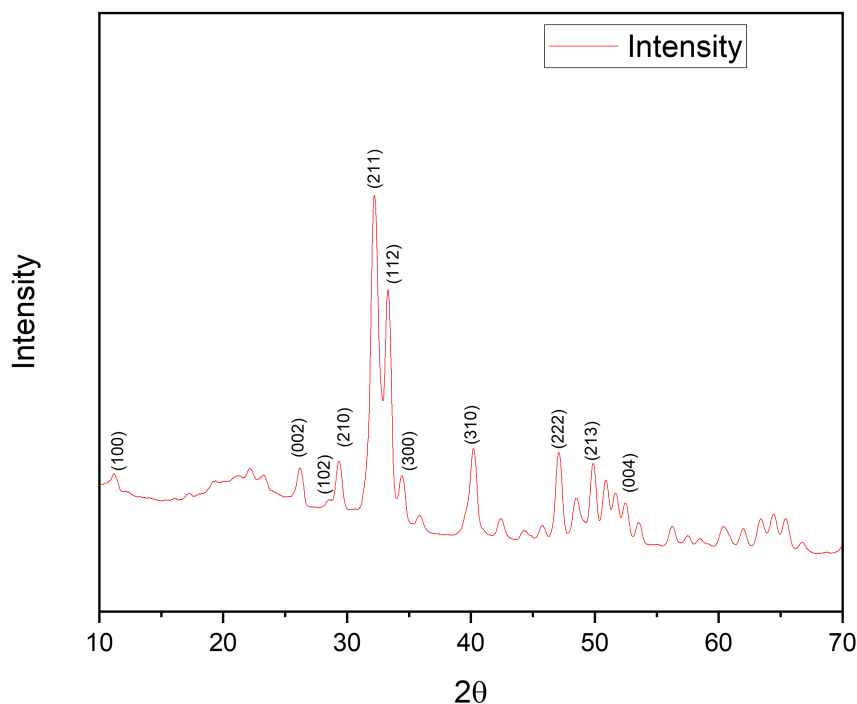


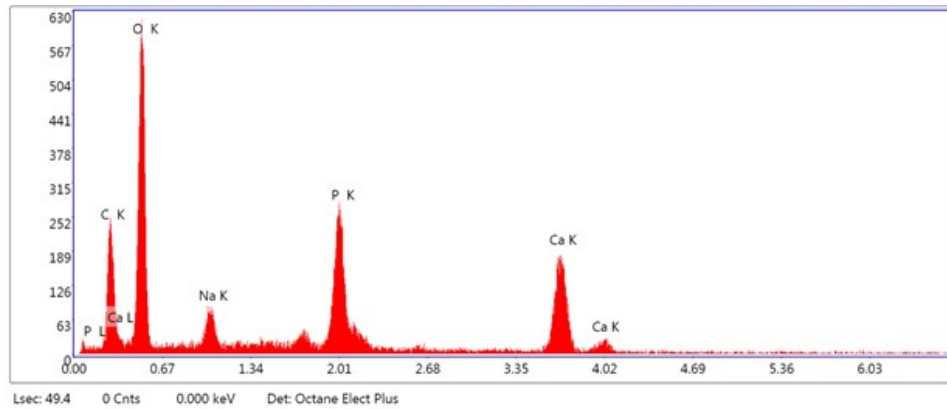
Fig 7: XRD patterns of ultralong HAP nanowires synthesized by the calcium oleate precursor solvothermal method using Oleic Acid

The EDS spectrum of the HAP nanowires reveals the elemental composition of the sample as shown in Fig 8. The spectrum shows prominent peaks corresponding to oxygen (O), carbon (C), calcium (Ca), phosphorus (P), and sodium (Na).

The peak at around 0.52 keV is attributed to oxygen (O). The peak at around 0.28 keV corresponds to carbon (C), which may be present due to organic contaminants or carbon-based precursor residues. The peaks at around 1.00 keV (Ca) and 4.02 keV (Ca) are indicative of calcium, a primary constituent of hydroxyapatite. - The peak at around 2.01 keV is assigned to phosphorus (P), another primary component of hydroxyapatite. - The presence of a peak at around 1.04 keV (Na) suggests a small amount of sodium in the sample.

The detected elements (Ca, P, O) confirm the formation of hydroxyapatite (HAP) nanowires, as these elements are consistent with the expected composition of HAP ($\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$). The presence of carbon could be due to residual organic molecules

from the precursor or adsorbed atmospheric CO₂.



(a)

Fig 8: Energy-Dispersive X-ray Spectroscopy (EDS) results for the HAP nanowires synthesized by the calcium oleate precursor solvothermal method using oleic acid.

Table 3: Elemental Composition from EDS Analysis of HAP Nanowires

Element	Weight %	Atomic %	Error %
C K	12.63	22.57	11.90
O K	33.10	44.41	10.09
Na K	3.20	2.99	12.79
P K	17.04	11.81	4.73
Ca K	34.02	18.22	5.98

EDX Calculations

Calculation of Ca/P Ratio

Based on Weight Percentage:

Weight % of Ca = 34.02%

Weight % of P = 17.04%

$$\text{Ca/P (weight)} = \frac{\text{Weight \% of Ca}}{\text{Weight \% of P}} = \frac{34.02}{17.04} \approx 1.996$$

Based on Atomic Percentage:

Atomic % of Ca = 18.22%

Atomic % of P = 11.81%

$$\text{Ca/P (atomic)} = \frac{\text{Atomic \% of Ca}}{\text{Atomic \% of P}} = \frac{18.22}{11.81} \approx 1.543$$

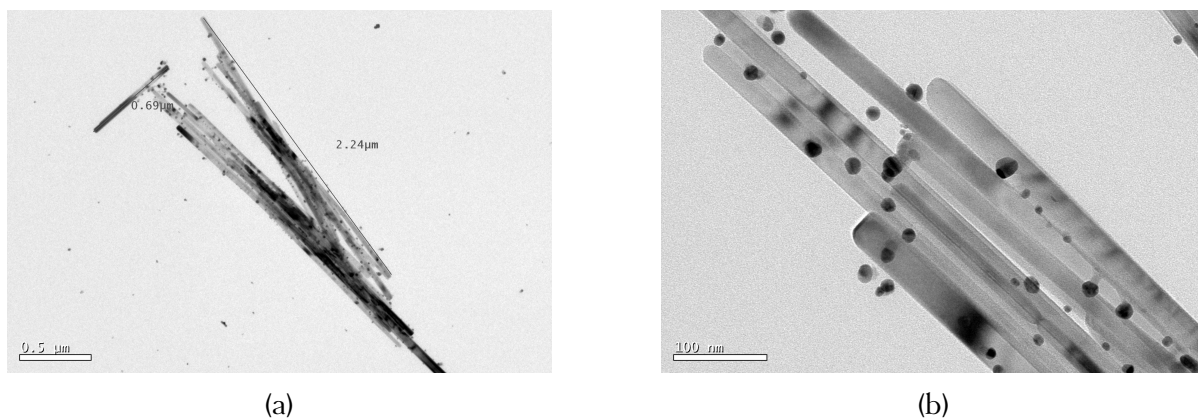
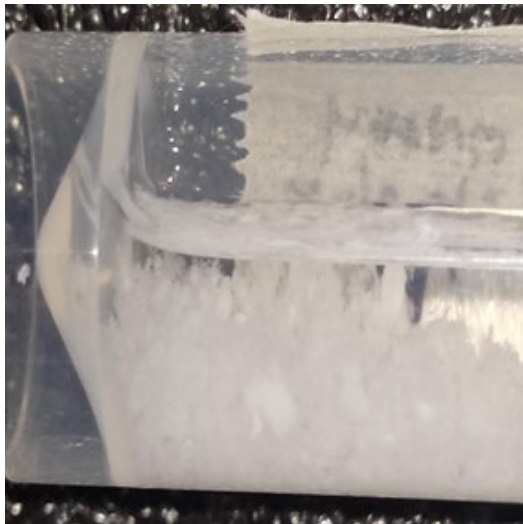


Fig 9 : (a) TEM micrograph of Ag-Doped was observed at $0.5\mu\text{m}$, (b) TEM micrograph of Ag-Doped was observed at 100nm

Conclusion

In this study, we successfully synthesized ultralong hydroxyapatite (HAP) nanowires using different precursors and characterized them through various analytical techniques.

The SEM analysis confirmed the formation of nanowires with a uniform diameter, indicative of a controlled synthesis process. TEM of Ag-Doped HAPNW was observed as shown in Fig 9. We observed Ag nanoparticle clumping on the HAP nanowire. FTIR spectra revealed the presence of functional groups associated with both the HAP and the organic precursors, while XRD patterns demonstrated the high crystallinity and hexagonal structure of the synthesized HAP nanowires. EDS analysis further confirmed the elemental composition consistent with hydroxyapatite, including calcium, phosphorus, and oxygen. As shown in the Fig 10, We obtained three distinct samples, a slurry-like liquid paste which we was obtained after the centrifugation steps, a fibrous HAP nanowire dispersed in Ethanol in which we only washed it with ethanol and removed the excess oil, and silver-doped HAP filtered and obtained in powder form. The fibrous HAP nanowires, when dispersed in ethanol, displayed fiber-like properties, suggesting their potential application in bioink synthesis. Our findings indicate that washing the HAP nanowires with ethanol without centrifugation is crucial to maintaining their integrity and fiber-like properties. This method ensures the appropriate calcium-to-phosphate ratio, leading to the formation of complete HAP nanowires suitable for bone tissue engineering applications.



(a)



(b)



(c)

Fig 10 : (a) Ethanol dispersed HAPNW without centrifugation (b) Slurry obtained with centrifugation steps (c) Ag doped Sample with grey colour

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